Chidubem Emeka-Nwuba

Email: dubem1407@gmail.com | Cell : (639)-994-7440 | LinkedIn: linkedin.com/in/chidubem-emeka-nwuba

EDUCATION

Bachelor of Science in Engineering - Computer Engineering **2022 - 2026**
University of Saskatchewan, Saskatoon, SK.
Graduated: April 2026
Current Program Average: 80% | Dean's Honor Roll: awarded in 1st, 2nd and 3rd Year.
Relevant Coursework: Embedded Systems | Digital Systems Architecture | System Verification & Testing | Control Systems | Digital-Analog Converters

HIGHLIGHTS OF SKILLS AND ABILITIES

- Knowledge of instrumentation, control systems, embedded systems, and system integration
- Strong technical troubleshooting, testing, and problem-solving abilities
- Experience working with hardware/software systems, signal processing, and real-time systems
- Proficient in C, Python, Bash, SystemVerilog, Linux, Git, Quartus, ModelSim, and VS Code
- Strong communication and collaboration skills working within multidisciplinary teams

PROJECTS

Obstacle Avoidance Subsystem Engineer

Mobile Tool Assistant (Capstone Project) | Team Project | *2025 - Present*

- Led development of the obstacle detection subsystem for an autonomous mobile robot designed to follow workers and transport tools, designing and implementing sensor-based avoidance logic and integrating it with embedded control software, resulting in reliable real-time obstacle response during system testing.
- Collaborated with localization and motion control teams to integrate sensor inputs into the broader control architecture, debugging hardware-software interactions and improving system stability during multi-module testing.

Digital Design Engineer

Traffic Light Controller | Course Project | *2024*

- Developed a traffic light control system with pedestrian and left-turn logic, designing state machines in SystemVerilog and implementing full control sequencing, resulting in verified functional behavior under all defined traffic scenarios.
- Created structured testbenches in Questa to validate timing and logical correctness, identifying and resolving edge-case failures prior to final synthesis.

Embedded Systems Developer

RTOS for Simulated Car | Course Project | *2024*

- Developed a real-time control system using µC/OS-II for a simulated vehicle platform, implementing task scheduling, inter-task communication, and synchronization mechanisms to meet deterministic timing constraints.
- Debugged race conditions and timing inconsistencies during integration, stabilizing system performance under concurrent task execution.

WORK EXPERIENCE

Audiovisual and Lighting Technician **2023 - Present**
TAG AVL Productions

- Supported execution of 50+ live events annually, configuring and troubleshooting complex audio and video systems in time-sensitive environments.
- Reduced setup time by approximately 15–20% by improving equipment organization and standardizing configuration workflows.
- Diagnosed and resolved signal flow and hardware issues during live productions, minimizing disruptions and maintaining event schedules.

ADDITIONAL SKILLS

- **Programming:** C, Python, Java, Bash, SQL
- **Tools:** Git, Linux, Quartus, ModelSim, VS Code
- **Concepts:** Testing & Debugging | Embedded Systems | Control Systems | Hardware Integration

References: Provided upon Request